

THA WANG PHA, Thailand

WMO: 483150

Lat: 19.1231N

Lon: 100.8131E

Elev: 776

StdP: 14.29

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.2	56.0	50.4	55.9	59.4	53.0	61.5	60.9	9.6	80.4	7.7	81.0	0.0	N/A	0.279

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	23.1	99.9	78.3	97.7	78.2	95.6	78.0	82.7	92.8	81.8	91.6	81.1	90.3	2.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	161.0	88.5	79.4	156.9	87.3	78.8	153.8	86.3	47.5	93.2	46.5	91.9	45.6	90.8	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.8	7.2	6.2	DB	48.5	103.2	3.4	2.6	46.1	105.0	44.1	106.5	42.2	108.0	39.7	109.8
			WB	47.9	84.3	3.3	1.5	45.5	85.4	43.6	86.2	41.7	87.0	39.3	88.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.7	71.2	75.5	80.3	84.7	84.4	83.8	82.4	81.8	82.2	80.8	76.8	71.8
	DBStd	5.51	3.75	3.30	3.78	3.20	3.25	2.63	2.60	2.13	2.06	2.37	3.57	4.24
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	10	5	0	1	0	0	0	0	0	0	0	0	4
	CDD50	10828	658	714	939	1042	1068	1015	1003	987	966	956	804	676
	CDD65	5364	198	294	475	592	603	565	538	522	516	491	354	216
	CDH74	53906	1556	2837	5133	6970	6778	6352	5533	5056	4996	4310	2797	1588
	CDH80	22787	453	1343	2849	3916	3242	2619	1923	1577	1750	1632	1041	442
(14) Wind	WSAvg	0.7	0.5	0.8	0.9	0.9	0.9	1.1	1.0	0.7	0.5	0.4	0.4	0.4
(15) Precipitation	PrecAvg	59.1	0.6	0.4	1.8	4.0	7.0	7.0	11.2	12.8	9.3	3.1	1.1	0.6
	PrecMax	79.0	2.6	2.4	5.9	7.4	14.7	15.4	19.6	21.8	15.8	6.6	5.8	3.5
	PrecMin	47.5	0.0	0.0	0.0	0.8	2.3	3.0	6.8	6.4	5.5	0.7	0.0	0.0
	PrecStd	8.9	0.7	0.6	1.5	1.7	2.6	2.8	2.9	4.0	2.8	1.4	1.3	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.1	95.6	100.4	103.8	102.4	97.8	96.1	92.9	93.4	93.4	91.8	89.1
		MCWB	72.1	73.0	75.7	78.5	80.8	81.3	80.7	80.9	80.9	79.3	76.8	74.2
	2%	DB	86.9	93.2	98.0	100.9	98.3	94.9	92.6	90.8	91.5	91.2	89.7	86.6
		MCWB	71.5	73.1	75.2	78.3	80.6	80.8	80.7	80.5	80.3	78.6	76.0	73.2
	5%	DB	84.9	91.1	96.0	98.5	95.5	92.6	90.3	89.0	90.0	89.7	88.1	84.6
		MCWB	70.9	72.8	74.9	78.6	80.2	80.4	80.1	80.0	79.7	78.2	75.5	72.1
	10%	DB	82.7	88.6	93.7	96.1	92.9	90.4	88.2	87.2	88.4	88.1	86.3	82.5
		MCWB	70.1	72.4	74.8	78.7	79.9	79.9	79.6	79.5	79.3	77.6	74.7	70.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.8	79.0	81.1	83.5	83.9	83.3	82.9	82.4	82.6	81.5	79.2	77.1
		MCDB	83.4	89.9	94.9	95.0	95.0	93.3	92.1	90.2	90.7	89.2	87.4	85.0
	2%	WB	73.8	76.0	78.8	81.9	82.5	81.9	81.6	81.2	81.3	80.2	78.1	75.5
		MCDB	82.7	88.1	93.0	95.0	93.2	91.5	89.9	88.6	89.1	88.4	86.5	83.1
	5%	WB	72.4	74.6	77.3	80.8	81.4	81.0	80.7	80.5	80.5	79.3	77.2	74.0
		MCDB	81.2	87.0	91.3	93.9	92.0	90.0	88.4	87.4	88.1	87.2	85.1	81.4
	10%	WB	71.1	73.2	76.0	79.6	80.5	80.3	79.9	79.7	79.8	78.4	76.1	72.6
		MCDB	79.7	85.4	89.7	92.7	90.8	88.5	86.9	86.0	86.7	85.9	83.5	79.9
(35) Mean Daily Temperature Range	5% DB	MDBR	24.1	28.7	27.5	23.1	16.9	13.0	11.3	11.1	13.0	15.6	19.2	22.1
		MCDBR	27.0	31.8	31.6	26.9	21.2	16.7	14.8	14.0	15.3	17.3	20.4	23.2
		MCWBR	12.6	13.4	11.4	8.6	6.9	5.6	5.6	5.5	5.7	6.4	8.3	10.6
	5% WB	MCDBR	22.8	27.9	27.2	24.1	18.7	14.9	13.6	12.9	14.4	15.9	17.8	19.8
		MCWBR	11.5	12.9	11.1	9.8	7.3	5.8	5.7	5.6	6.0	6.5	8.1	9.6
(40) Clear-Sky Solar Irradiance	taub		0.448	0.555	0.727	0.692	0.506	0.452	0.448	0.468	0.494	0.534	0.434	0.432
	taud		2.170	1.886	1.543	1.664	2.169	2.349	2.352	2.277	2.185	2.032	2.278	2.239
	Ebn at Noon		261	239	204	214	255	267	268	265	257	241	264	262
	Edh at Noon		44	62	91	81	49	40	40	43	47	53	40	41
(44) All-Sky Solar Radiation	RadAvg		1400	1629	1685	1830	1669	1492	1370	1360	1443	1442	1449	1330
	RadStd		83	86	98	123	153	114	117	88	65	68	94	82

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+274	+270

Nomenclature: See separate page